

Q1. Consider the following model: $Y_i \sim \text{Ber}(p)$, $Z_i \sim N(0,1)$, $X_i \sim N(W_k Z_i, s)$ where $k = Y_i$. We have 3 observations of X and Y : $[X_1=3.6 \ Y_1=1, X_2=-1.2 \ Y_2=0, X_3=4.2 \ Y_3=1]$. (Note that we do not observe Z .) Consider the following 2 choices of parameters:

P1: $[W_0=-5, W_1=5, s=9, p=0.5]$, P2: $[W_0=0, W_1=4, s=25, p=0.8]$

Which of these two sets of parameters fits the observations best?

Q2. Consider a 2-state Hidden Markov Model with initial state distribution $\pi=\{0.4,0.6\}$, and state transition matrix $T= \begin{bmatrix} 0.8 & 0.2 \\ 0.1 & 0.9 \end{bmatrix}$. It emits two discrete values A and B, and the discrete emission matrix is $E=\begin{bmatrix} 0.7 & 0.3 \\ 0.1 & 0.9 \end{bmatrix}$. Suppose it emits an output sequence $X=[A,A,B]$. What is the probability that the first hidden state $Z_1= 1$?

Q3. Consider a short sequence of observations $[3.6 \ 7.5 \ 2.4]$. Based on AIC, which of the following models would you choose to fit these observations?

M1: a GMM with 2 states with $\pi=[0.6, 0.4]$, $\mu=[3, 9]$, $\sigma^2=[1, 4]$

M2: a HMM with 2 states with $\pi=[0.5, 0.5]$, $A=\begin{bmatrix} 0.6 & 0.4 \\ 0.4 & 0.6 \end{bmatrix}$, $\mu=[3, 7.5]$, $\sigma^2=[1, 1]$

According to the chosen model, calculate the probability that the second observation (7.5) was generated by the second state, i.e. $Z(2)=2$.

Q4. Consider a Hidden Markov Model with 3 possible latent state values, with Gaussian emission distribution with parameters $(-3, 9)$, $(0, 4)$, $(3, 9)$. The state transition parameters follow the relation: $p(Z(t)=j | Z(t-1)=i) \propto 1/(|i-j|+1)$. All states are equally likely at $t=1$. You are provided with a short sequence of observations $X=(0.8, -2, 1.5)$. Find the probability distribution of $Z(3)$, using the observations and parameters.

Q5. You are given 4 sequences of length 6, where both cluster variables and observations are provided in the format $Z(t)/X(t)$. The observations follow a Gaussian distribution, conditioned on the cluster variables, i.e. $X(i) \sim N(u(k), s(k))$ where $k = Z(i)$.

- (i) Estimate if the data follows GMM or HMM with Gaussian emissions. Use Maximum Likelihood for parameter estimation.
- (ii) You are now provided a news test sequence $[X(1), \dots, X(4)]$. Compute the joint distribution of $[X(5), X(6)]$, based on the model parameters estimated in part (i).

S1: $1/-0.5, 1/-0.8, 2/1.3, 3/5.0, 3/4.7, 2/1.1;$

S2: $1/-0.5, 1/0.4, 1/0.6, 2/0.7, 3/5.2, 3/5.3;$

S3: 2/1.1, 3/4.5, 3/5.2, 3/4.9, 2/0.9, 3/5.1;

S4: 1/0.5, 2/1.2, 1/-0.4, 1/0.7, 2/0.7, 3/4.6;

TEST: $\frac{1}{1.0}$, $\frac{1}{4.9}$, $\frac{1}{5.1}$, $\frac{1}{1.1}$,

Q6. Consider the following latent variable model: $Z_i \sim \text{Cat}(\pi)$, $Y_i \sim U(a_k, b_k)$, $X_i \sim N(Y_i, v_k)$ where $k = Z_i$. Clearly, the model parameters are $[\pi, \{a, b, v\}_k]$. The observations are $[x_1, x_2, \dots, x_N]$ (all real numbers), while Y, Z are latent. Y are real-valued while $Z \in \{1, 2, \dots, K\}$.

- i) Write the expression for the model log-likelihood
- ii) Discuss how the parameters may be estimated using E-M algorithm. Explain what will be done in the E-step and M-step.
- iii) Derive $p(Y_i | X_i, Z_i)$ and $p(Z_i | X_i, Y_i)$ in terms of the parameters
- iv) Suppose Y, Z are known along with X . Derive ML estimates of all parameters